

System Advisor Model Report

Detailed Photovoltaic
Commercial

314 DC kW Nameplate
\$1.86/W Installed Cost

38.89, -77.02 UTC -5
NSRDB

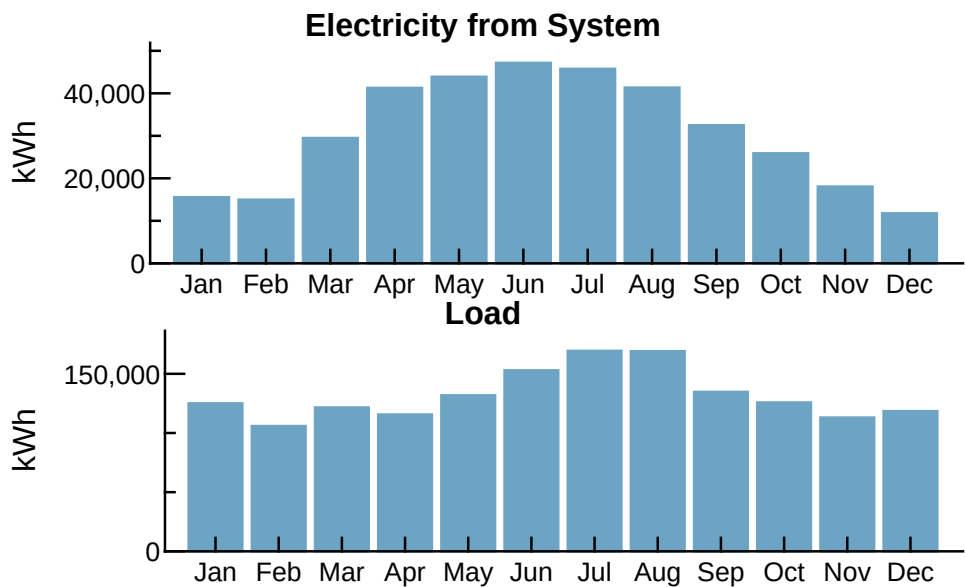
Performance Model			Financial Model	
Modules			Project Costs	
Aptos Solar Technology LLC DNA-144-BF10-530W			Total installed cost \$583,665	
Cell material	Mono-c-Si Bifacial		Salvage value \$0	
Module area	2.58 m²		Analysis Parameters	
Module capacity	530.75 DC Watts		Project life 25 years	
Quantity	591		Inflation rate 2.5%	
Total capacity	313.67 DC kW		Real discount rate 5%	
Total area	1,524 m²		Project Debt Parameters	
Inverters			Debt fraction 0%	
SMA America: STPS27US-20			Amount \$0	
Unit capacity	27 AC kW		Term 25 years	
MPPT Voltage Range	200 - 980 VDC		Rate 7%	
Nominal AC Voltage	208.0 VAC		Tax and Insurance Rates	
Quantity	10		Federal income tax 0 %/year	
Total capacity	270 AC kW		State income tax 0 %/year	
DC to AC Capacity Ratio	1.16		Sales tax (% of indirect cost basis) 13%	
AC losses (%)	0.00		Insurance (% of installed cost) 0 %/year	
Two subarrays:			Property tax (% of assessed val.) 0 %/year	
	1	2	Incentives	
Strings	23	21	None	
Modules per string	12	15	Electricity Usage and Rate Summary	
String Voc (DC V)	590.40	738.00	Annual peak demand 597.8 kW	
Tilt (deg from horizontal)	20.00	20.00	Annual total usage 1,583,134 kWh	
Azimuth (deg E of N)	180	0	Generic Commercial	
GCR	0.3	0.3	Fixed charge: \$125.96/month	
Tracking	no	no	Monthly excess with kWh rollover	
Backtracking	-	-	Flat energy buy rate: \$0.145/kWh	
Self shading	yes	yes	Results	
Rotation limit (deg)	-	-	Nominal LCOE 18.5 cents/kWh	
Shading	no	no	Net present value \$61,500	
Snow	no	no	Payback period 10.5 years	
Soiling	yes	yes		
DC losses (%)	4.44	4.44		
Performance Adjustments				
DC avail./curtail.	2%			
AC avail./curtail.	2%			
Degradation	0.5 %/yr			
Hourly or custom losses	none			
Annual Results (in Year 1)				
GHI kWh/m²/day	4.28	4.28		
POA kWh/m²/day	110.00	82.00		
Net to inverter	399,194 DC kWh			
Net to grid	367,656 AC kWh			
Capacity factor	13.4			
Performance ratio	0.79			

Detailed Photovoltaic
Commercial

314 DC kW Nameplate
\$1.86/W Installed Cost

38.89, -77.02
UTC -5

Year 1 Monthly Generation and Load Summary



Year 1 Monthly Electric Bill and Savings (\$)

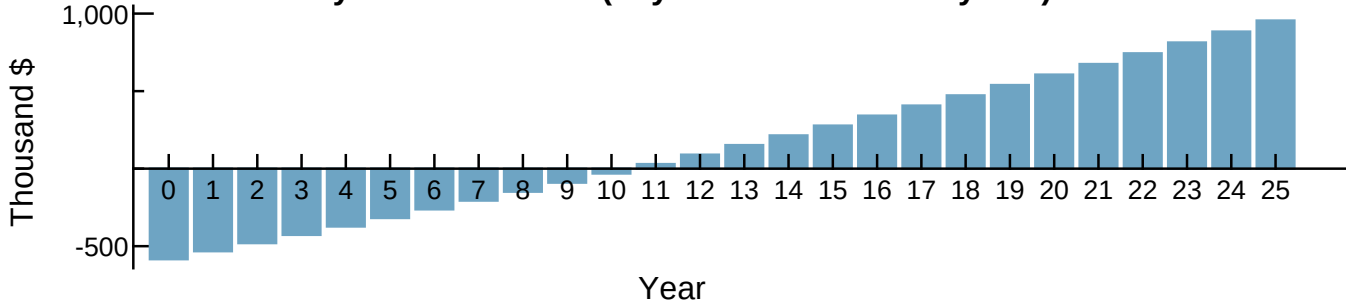
Month	Without System	With System	Savings
Jan	24,226	21,974	2,251
Feb	19,890	17,719	2,170
Mar	22,128	17,260	4,867
Apr	21,758	14,760	6,997
May	24,286	17,310	6,976
Jun	28,322	20,417	7,904
Jul	31,672	24,203	7,468
Aug	31,269	25,166	6,103
Sep	25,432	20,276	5,156
Oct	23,076	19,103	3,972
Nov	20,491	17,663	2,827
Dec	21,988	20,285	1,703
Annual	294,543	236,143	58,399

NPV Approximation using Annuities

Annuities, Capital Recovery Factor (CRF) = 0.0907		
Investment	\$-52,900	Sum:
Expenses	\$-10,900	\$5,500
Savings	\$0	NPV = Sum / CRF:
Energy value	\$69,500	\$61,000

Investment = Installed Cost - Debt Principal - IBI - CBI
Expenses = Operating Costs + Debt Payments
Savings = Tax Deductions + PBI
Energy value = Tax Adjusted Net Savings
Nominal discount rate = 7.625%

Payback Cash Flow (Payback Period = 10.5 years)



Detailed Photovoltaic
Commercial

314 DC kW Nameplate
\$1.86/W Installed Cost

38.89, -77.02
UTC -5

